



Foundations of Programming

Assignment No. : 9

PROBLEM STATEMENT:

Write a program in C to check whether the given number is an Armstrong number or not. An Armstrong number is an integer with three digits such that the sum of the cubes of its digits is equal to the number itself.

Example: 371

OBJECTIVE:

- To understand the concept of Armstrong numbers.
- To practice arithmetic operations in C.
- To develop logical thinking using loops and conditional statements.

THEORY:

An Armstrong number (also called a narcissistic number) is a number that is equal to the sum of the cubes of its digits. For a three-digit number abc , it is an Armstrong number if:

$$a^3 + b^3 + c^3 = abc$$

For example, 371 is an Armstrong number because:

$$3^3 + 7^3 + 1^3 = 27 + 343 + 1 = 371.$$

PLATFORM:

Linux

ALGORITHM:

Step 1: Start

Step 2: Declare variables

num, original, digit, sum = 0

Step 3: Accept a number from the user.

Step 4: Store the number in original.

Step 5: Repeat while $\text{num} > 0$

a) Find last digit $\rightarrow \text{digit} = \text{num} \% 10$

b) Add cube of digit to sum $\rightarrow \text{sum} = \text{sum} + \text{digit}^3$

c) Remove last digit $\rightarrow \text{num} = \text{num} / 10$

Step 6: Compare sum with original

- If equal → Display “**Armstrong number**”
- Else → Display “**Not an Armstrong number**”

Step 7: Stop

C PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    int num, original, digit, sum = 0;
```

```
    printf("Enter a 3-digit number: ");
```

```
    scanf("%d", &num);
```

```
    original = num;
```

```
    while (num > 0) {
```

```
        digit = num % 10;
```

```
        sum = sum + digit * digit * digit;
```

```
        num = num / 10;
```

```
    }
```

```
    if (sum == original)
```

```
        printf("Armstrong number");
```

```
    else
```

```
        printf("Not an Armstrong number");
```

```
    return 0;
}
```

```
1  #include <stdio.h>
2
3  int main() {
4      int num, original, digit, sum = 0;
5
6      printf("Enter a 3-digit number: ");
7      scanf("%d", &num);
8
9      original = num;
10
11     while (num > 0) {
12         digit = num % 10;
13         sum = sum + digit * digit * digit;
14         num = num / 10;
15     }
16
17     if (sum == original)
18         printf("Armstrong number");
19     else
20         printf("Not an Armstrong number");
21
22     return 0;
23 }
24
25
```

INPUT:

Enter a three-digit number.

Example Input:

371

OUTPUT:

```
Enter a 3-digit number: 371
Armstrong number
```

CONCLUSION:

Thus, we have successfully written a C program to check whether a given number is an Armstrong number using loops and arithmetic operations.

FAQs:

1. What is an Armstrong number?
2. Why is the Armstrong number concept important in programming?
3. How does this program work step by step?